any like this: a physical quantity has a certain value." He also claimed that Bohr's work supported this "dual description" of nature. Yet von Neumann's view on quantum interpretation certainly didn't line up with Bohr's. Indeed, there was a wide gap between Bohr and the "Martians" not only on wave function collapse and the application of quantum theory to measurement devices, but also on complementarity. Wigner had disparaged complementarity when Bohr had first unveiled the idea at Como in 1927, and von Neumann had made little use of the idea in his textbook. Now that von Neumann and others were questioning the Copenhagen orthodoxy on several fronts, a confrontation over the foundations of quantum theory seemed likely.

But by the late 1930s, Bohr, von Neumann, and Wigner had little time to think about the foundations of quantum physics. War was obviously on the horizon, and new developments in more practical branches of physics overtook concerns about the philosophical underpinnings of the field. In January 1939, Bohr and his assistant Léon Rosenfeld took a steamer across the Atlantic, bearing to Manhattan the latest news from the Continent: the German physicist Otto Hahn had split the atom. Bohr immediately tackled the problem. With the help of his former student John Wheeler, the father of quantum physics set about unveiling the mysteries of uranium.

The phenomenal power of atomic bombs ultimately derives from the delicate balancing act that is performed in the nucleus of every atom. The cloud of electrons surrounding an atomic nucleus is bound to the nucleus by the electric attraction between the negatively charged electrons and the positively charged protons in the nucleus. But that same electric force also tries to rip the nucleus apart—like charges repel, and the closer together they are, the more they repel. A typical atomic nucleus is 100,000 times smaller than the surrounding electron cloud, which is itself a million times smaller than the width of a human hair. At such close quarters, the electrical repulsion between the protons in the nucleus, left unchecked, would send them flying off at nearly the speed of light. Instead, atomic nuclei are held together by an even stronger force, unimaginatively dubbed the "strong